

Governance Patterns in Global Software Engineering: Best Practices and Lessons Learned

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Abstract—Over the past decade Global Software Engineering (GSE) has become the norm influenced by several factors such as market evolutions in emerging economies, availability of skilled pool of resources at optimal costs and globalization trends such as mergers and acquisitions. Evidently, market leaders, large businesses as well as niche players in Software Product Engineering have leveraged GSE in several areas such as product development, sustenance, support, testing and professional services. In this context, governance of GSE projects or engagements is a critical factor that provides adequate foundation and support structure to ensure results. This paper discusses three fundamental governance patterns and additional derived patterns observed in a set of GSE projects. Also, this paper presents two case studies and our experiential findings on governance perspective.

I. INTRODUCTION

The vision of GSE for any given project is to deliver work products of high quality on schedule by means of engaging geographically distributed teams that define the right architecture for distributed development and to follow the right processes and tools for coordination and communication [6]. There are several challenges encountered in areas such as Requirement Engineering, Change Management and Project Management to name a few [6]. However, practitioners continue to evolve methodologies such as Distributed Agile Software Development based on Agile Principles [4, 8] in order to ensure value delivery [4] in distributed projects. The governance [1] of GSE projects and engagements plays a significant role in doing the right projects in distributed model, doing them right time after time [11] and eliminating project failures during initial stages in order to improve the success rate of GSE projects. Besides, effective governance is required in order to manage risks, constraints and issues related to a range of factors including legislative, competitive, and cultural changes applicable to organizations across the globe [11].

The word governance originated from the Greek verb κυβερνάω (kubernáo) which means ‘to steer’. It was used for the first time in a metaphorical sense by Plato. In general governance means a mechanism that includes a group of people (or committees or departments etc.) who make up a body for the purpose of administering something

and hence to make the best decisions in a timely manner. In case of software projects executed at a single location it has been a general practice to implement a governance mechanism at three levels namely project level, program level and organizational level [2, 3, 10]. In case of GSE projects executed across multiple geographies and time zones with the employees of project sponsor organization, external vendors and independent contractors the complexity of governance increases multifold. With more than a decade of work experience in GSE, we have identified three fundamental patterns of GSE governance that give rise to additional derived patterns. GSE engagements typically start with a governance mechanism that involves one of these patterns and mature towards governance patterns that yield better results or provide a better fitment to the overall strategy of GSE initiatives.

II. GOVERNANCE PATTERNS

In any organization a team that focuses on governance activities that are strategic in nature involves the senior leaders of the organization. At project level or program level even though the project manager or program manager manages the project or program, at GSE context it is necessary to have a governance structure that includes the executive sponsor [2] and senior leaders across all locations. This enables globally distributed teams in aligning themselves with the GSE initiative as conceived jointly by the governance team. Structurally governance patterns are very similar to engagement patterns. However in GSE context the term governance pattern fits well when we consider the structural as well as behavioral elements. In any GSE project the governance mechanism drives factors such as performance management of workflow processes and service levels. Besides, the responsibility of financial management, contract management, relationship management and resource management fall under the scope of governance. Governance teams setup for local teams need to form a GSE governance team and work together as a single body at global level in order to run GSE projects and engagements successfully.

In general, GSE governance teams schedule periodic steering committee meetings to review the status of GSE initiatives and arrive at action items. These meetings are attended by governance team members across geographies through conference calls or video conferencing. Optionally, once in a year or every six months governance team

members plan to meet at a central location to have face-to-face review meetings.

The three fundamental governance patterns observed in GSE context are Delegate, Collaborate and Multi-Partner. From these patterns additional patterns can be derived to suit specific implementation of GSE governance.

A. Delegate

In this pattern (Fig. 1) the parent entity or project sponsoring organization (C) delegates work to its remote entity widely known as captive center in another location (C). Subsequently, the remote entity delegates the entire work or part of the work to in-house teams or one or more service providers (P1). In Fig. 1 through Fig. 4, the solid lines show the direction of work delegation and the dotted lines show the communication paths of GSE governance team members.

Typically in this pattern the senior management of the parent entity delegates GSE decisions to the remote middle management (Directors and Managers) with approved budgets. Project governance becomes the responsibility of the remote entity. Steering committee reviews involving stakeholders from the parent entity are done on need basis only. These reviews happen once or twice every year to focus on short-term goals that are not strategic.

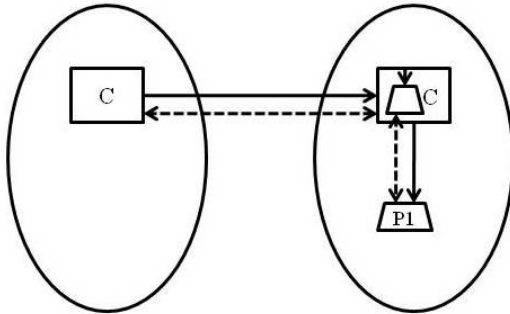


Figure 1. Delegate

The benefit of this pattern as seen in several instances emerges from the complete ownership of the remote captive center. When this happens, the parent entity participates on need basis and waits for finished product or end results. While this is an encouraging factor for the remote entity, this pattern may impede projects because of the dependency of remote entity on parent entity in situations that involve collective decision making on critical matters such as additional funding or negotiating project delays. Also, the absence of continuous collaboration on understanding the engineering processes, methods and protocols may cause disconnect among parent and local entities [3]. In this model, service providers engaged with the local entity are limited to providing augmented staffing services or participating in small projects. Because of this, they seldom get a big picture on their work responsibility as well as long-term visibility on product roadmap. Tactical measures in

engaging service providers cause challenges in knowledge retention as well as executing critical projects.

In majority of instances this pattern is seen in its simplest form when the parent entity delegates work to a remote service provider in the absence of a captive center. Typically, projects executed in such instances are selected so as to mitigate IP violation risks and to experiment GSE before considering strategic or key projects. Depending on whether the service provider is located in the same country or off the shore this phenomenon is termed as outsourcing or offshoring. Most of the GSE initiatives originate with this governance pattern.

B. Collaborate

In this pattern (Fig. 2) teams spread across multiple geography collaborate to implement governance. The parent entity and remote entities or captive centers form project teams and implement collaborative governance. This pattern does not involve any outsourcing.

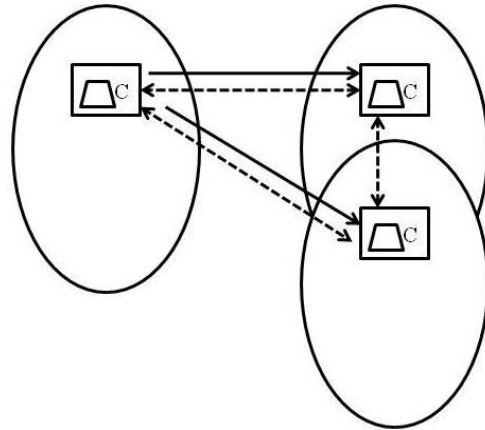


Figure 2. Collaborate

This pattern is employed in projects that are very complex in nature and involve confidentiality as well as protection of IP within designated corporate groups. Though this pattern constrains project sponsor from utilizing the strengths and maturity of service providers, it serves the purpose in executing strategic projects such as new product development initiatives. The governance team includes leaders from all locations and hence this pattern involves periodic meetings and decision making opportunities to avoid delays during project execution.

A variation of this pattern can be observed when an outsourced work is done across multiple cities in a country by a service provider (Fig. 3). In this case, service provider implements governance mechanism locally and rolls it up to the executive sponsor in order to implement collaborative GSE governance. In this situation, it is necessary to form a steering committee that comprises of leaders from the parent organization as well as service providers.

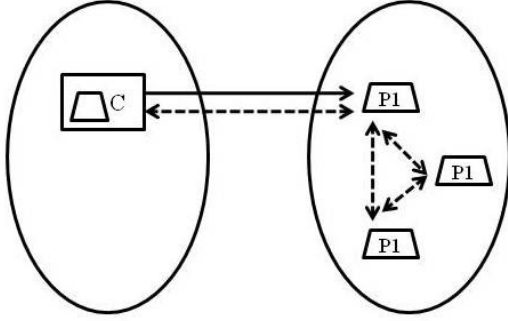


Figure 3. Collaborate – A Variation

C. Multi-Partner

In this pattern (Fig. 4) the parent organization works with multiple partners and optionally has one or more local entities or captive centers. This pattern fits well in case of large projects outsourced by the parent entity to multiple local entities or service providers in order to mitigate the risk of becoming dependent on a single entity. Also, it provides a mechanism to segment the outsourced work to negotiate optimal price points as well as service levels. For example, using this pattern, partitioning of product development or sustenance and product testing activities across partners provides an opportunity for independent testing and quality assurance. Also, partners get an opportunity to have visibility on long-term product roadmap of project sponsors and align themselves well enough to execute projects. This enables congruence among project team members across locations over a period of time [5].

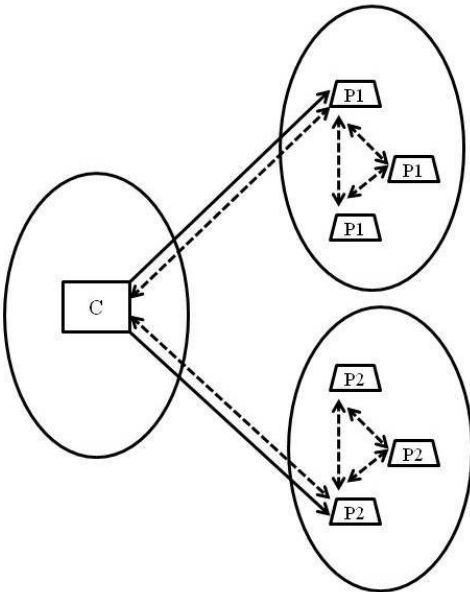


Figure 4. Multi-Partner

In practice, project sponsors implement this pattern by selecting a list of qualified service providers or vendors. Each qualified vendor executes designated projects of one or

more product lines. This variation can be termed as Qualified-Partner. In some instances of this pattern the implementation of IT security policies defined by project sponsors, commissioning of physically secured workspace as well as facilitation of annual planning and steering committee reviews become the salient factors.

Another variation of Multi-Partner is when project sponsors implement a mechanism to certify partners and involve only certified partners in GSE. This variation of Multi-Partner pattern can be termed as Certified-Partner. These two variations of Multi-Partner pattern nurture long-term relationships. Governance in Certified-Partner pattern provides for annual joint planning sessions, periodic steering committee reviews, commissioning of remote development centers [3] with IT security and policies as specified by project sponsors and assessment of IT infrastructure and security on a periodic basis.

In practice the adoption and implementation of GSE initiatives follow a specific governance pattern to begin with. This is necessary to collectively decide on a common set of governance parameters across sites to steer GSE projects towards success. Each governance pattern promotes a set of service offerings and relationships. As relationship among participating entities matures governance pattern also changes in order to accommodate revised strategies and goals. Typically, factors such as engagement maturity, product phase, service offering and cost model influence the maturity and transformation of governance patterns over a period of time.

III. CASE STUDY 1

This case study is about our experience in a GSE engagement with a multinational product development organization. This initiative started in 2007 with our participation as service provider with our customer's team in the US. During the first year we formed two teams of seventy engineers distributed across two cities in India. For our project sponsors cost effectiveness was the primary driver during the initial stages. Our customer facilitated a 2-month onsite knowledge transfer to some of the key team members of virtual teams. We adopted a minor variation of the delegate pattern of GSE governance. The service offerings were product sustenance and product testing. This pattern consumed significant efforts from the customer as some of the senior leaders were involved in screening and induction of team members for remote teams. Also, some of the initial projects were transactional and tactical in nature. Monthly governance meetings were conducted over conference calls during the first few months. After knowledge transfer and stabilization phases we mutually agreed to facilitate governance meetings on a quarterly basis. Over a period of twelve months from the initiation of the first project, we found difficulties in the overall engagement as it was not enabling us in executing project phases from remote locations because of disparate processes and leadership styles.

In one of our governance meetings we discussed the importance of collaborative governance across sites and the need for engagement maturity. This provided us a platform

to improve results by means of identifying common goals and parameters to measure success and resolving issues across sites. Also, it reduced our customer's overheads in independently governing two separate sites. The governance team decided to entrust remote teams with the responsibility of screening and inducting new hires as well as managing team level issues. With this transition, we were able to accommodate new service offerings such as professional services and new product development. This helped us execute distributed projects that implemented agile principles and followed Scrum. Overall, transitioning into collaborate pattern helped us in strengthening local governance as well as engineering methodologies and practices followed by the virtual teams.

IV. CASE STUDY 2

We started this engagement during 2005 with a product development organization based in Europe which was acquired by a US based Independent Software Vendor (ISV) after few years. The GSE strategy of the customer was to have an independent testing team of 10 engineers based in India and conduct all product development and sustenance activities in Europe. Senior members from customer team visited our offshore facilities and enabled knowledge transfer. We adopted a minor variation of the delegate pattern of GSE governance in this project. During the first two years, our service offerings were limited to manual product testing. Project sponsors were satisfied as the remote team was able to mature in manual product testing and deliver good results. The focus of the governance team that comprised of senior leadership from both organizations was to meet project schedule and provide adequate process quality on manual testing. We built the culture of presenting defect analysis reports and trends to present process quality for product quality assurance. We facilitated quarterly governance meetings over conference calls to present and discuss these parameters. The results of the first two years of this GSE initiative met the expectations of project sponsors.

The offshore team became well-versed in manual testing and wanted to get involved in test automation. The motivation was to find an alternative in place of monotonous work content, gain expertise as well as optimize testing efforts. With the delegated pattern of GSE governance it was challenging for us to bring in such changes. This was the point when we collaborated with our customer in transforming GSE governance in our project. Our team demonstrated the benefits of automation by developing a proof of concept and collaborated with our customer in evaluating two tools for test automation. Meanwhile, our customer formed an additional virtual team in India. With these changes we moved into collaborative governance that provides for collaboration of senior leadership from all virtual teams. Eventually, our team underwent a training program on a preferred automation tool and contributed to test automation projects. In this engagement we noticed that collaborative governance contributed to an increased level of collaboration among team members across sites and hence contributed to improving product quality as well.

V. BEST PRACTICES

A. Establish Shared Vision

Establishing a shared vision on GSE initiatives across governance team members is very essential. This has to be a collaborative exercise supported by the executive sponsor. This helps the senior leadership at each location understand the vision and sensitize team members with the right context and shared vision. Without this, virtual team members tend to restrict themselves to transactional engineering activities without relating their work to the overall business needs of project sponsors.

B. Define Success Parameters

Even though every project needs well-defined milestones and goals, it is very critical to define success parameters at governance level. This helps GSE governance teams understand project success in terms of a common set of parameters. Without this step, GSE governance teams tend to focus on transactional issues and miss the big picture.

C. Ensure Early Success

While it is imperative to have a long-term view of the future, it is equally important to focus on early success. One way to accomplish this is to define success parameters beyond tested code. To make this happen, any GSE governance team must have strong, visible commitment to the success of GSE initiatives [3]. Having a one year roadmap and identifying milestones or events that can be measured against success every quarter is a way to ensure early success and mitigate risks.

D. Review and Track Actions

Periodic steering committee reviews are very essential to understand and improve GSE initiatives. Having a collective decision on specific, measurable, action items that are realistic and time-bound at the end of each review and systematically tracking them to closure ensures positive reinforcement in governance. During initial stages it is required to have these reviews every month and as soon as the first few early successes happen, the frequency of these reviews can be once in two months or once in a quarter.

E. Facilitate Annual Planning Sessions

Annual planning sessions are essential in case of large engagements that encompass multiple service offerings. These sessions provide a forum to review the service levels, commercial factors as well as best practices of GSE initiatives and help stakeholders align themselves for the future.

VI. LESSONS LEARNED

Establishing strong local governance ensures successful GSE governance [3, 7]. In one of our projects we found that during steering committee reviews, the GSE governance team had to spend significant efforts in discussing and resolving transactional issues that could have been resolved

earlier by remote teams. We analyzed this situation and identified measures to strengthen local governance by means of aligning the parameters as well as frequency of local reviews. This helped us resolve several issues locally and provided adequate focus during our GSE steering committee meeting in discussing issues that impact teams and projects at a higher level.

In another instance we realized that having technically savvy members in GSE governance team helps resolve technical issues that impact global teams. In one of our Fixed Price projects that experienced more than 30% of schedule overrun, we conducted frequent reviews with the GSE governance team. As some of the team members were technically savvy, they were able to understand and appreciate technical challenges faced by global teams in an objective manner. This helped us negotiate revised scope and schedule by reprioritizing use cases and find a win-win solution.

VII. CONCLUSION

Governance has been one of the key success factors in GSE projects and it is going to provide the necessary foundation and support for GSE in future as well. The three fundamental patterns and additional derived patterns discussed in this paper inherit certain key factors and differentiators. For any of these patterns to be successful both the structure and behavior need to synchronize with the overall vision of GSE initiatives. It is evident that countries and corporate entities from new geographies have started becoming a part of GSE. Hence, during the next decade, focus on GSE governance is necessary in order to promote the success of GSE projects. GSE governance is in its

nascent stage and holds immense potential for further research both in the academia and the industry.

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